Rout Dutta.

HW4

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Stochastic models. (40.520) [CTMC].

1.) A system has 2 identical machines.
- Each working machine has a failing rate of 2/m.
- For each machine, repairman fixes one machine with a repairing rate of 4/m.

(A) Let the states be: '0', '1', '2' failed machine

$0 \longrightarrow 1$

Rate : 2 + 2 = 4.

$1 \longrightarrow 0$ (1 has 2 states $1 \to 0$ & $1 \to 2$)

Repair completion = 4.

$1 \longrightarrow 2$

Failure rate = 2.

$2 \longrightarrow 1$ (2 machine failed $\to$ 1 machine failed)

Repair rate = 4.

$\therefore$ $0 \longleftrightarrow 0$ (Net out rate = -4)

$0 \longrightarrow 1$ (4)

$0 \longrightarrow 2$ (0)

$1 \to 0$ (4)

$1 \to 1$ (Net out = (4)+(2) = -6)

$1 \to 2$ (2).

$2 \to 0$ (0)

$2 \to 1$ (4)

$2 \to 2$ (Net out = -4).

CTMC States
- (0 faulty) '0'
- (1 faulty, 1 good) '1'
- (2 faulty) '2'

$$\therefore [Q] = \begin{bmatrix} -4 & 4 & 0 \\ 4 & -6 & 2 \\ 0 & 4 & -4 \end{bmatrix}.$$

Generator Matrix

(b) Birth death CTMC

$\lambda_0 = 4$
$\lambda_1 = 2$
$\mu_1 = 4$
$\mu_2 = 4$

	Birth (failure)	Death (repair)	state
	4	–	0
	2	4	1
	–	4	2

$\pi_i \lambda_i = \pi_{i+1} \mu_{i+1}$.
$\pi_0 \lambda_0 = \pi_1 \mu_1$.
$\therefore \pi_0 . \cancel{4} = \pi_1 . \cancel{4} \rightarrow \pi_0 = \pi_1$.

$\pi_1 \lambda_1 = \pi_2 \mu_2$.

$\therefore \pi_1 . 2 = \pi_2 4$.

$\therefore \pi_2 = \pi_1/2 \rightarrow \pi_2 = \pi_1/2$.

$\pi_0 + \pi_1 + \pi_2 = 1$.
$\pi_0 + \pi_0 + \pi_0/2 = 1 \Rightarrow \frac{5\pi_0}{2} = 1$.
$\therefore \pi_0 = 2/5 \Rightarrow \pi_1 = 2/5$.
$\pi_2 = 1/5$

$\therefore E[\text{production}] = 2 . \pi_0 + 1 . \pi_1 + 0 . \pi_2$
$= 2 . 2/5 + 1 . 2/5 + 0$.
$= 6/5 = \boxed{1.2 \text{ units}}$

(i) Embedded chain.

$p_{ij} = \frac{q_{ij}}{-q_{ii}}$.

$\therefore p_{01} = 4/4 = 1$, $p_{10} = \frac{4}{6}$, $p_{12} = 2/6$, $p_{21} = 4/4$

$$\therefore P = \begin{bmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 0 & 1 & 0 \end{bmatrix}$$

To find π_i (or stationery probabilities):

$\therefore \pi = \pi \cdot P$.

$$\begin{bmatrix} \pi_0 & \pi_1 & \pi_2 \end{bmatrix} = \begin{bmatrix} \pi_0 & \pi_1 & \pi_2 \end{bmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\begin{bmatrix} \pi_0 & \pi_1 & \pi_2 \end{bmatrix} = \begin{bmatrix} \frac{2}{3}\pi_1 & \pi_0 + \pi_2 & \pi_1/3 \end{bmatrix}.$$

$\pi_0 = 2/3\,\pi_1$, $\pi_2 = \frac{\pi_1}{3} = \frac{\pi_0}{2}$.

$\pi_0 + \pi_1 + \pi_2 = 1 \Rightarrow \pi_0 + 3/2\,\pi_0 + \pi_0/2 = 1$.

$\Rightarrow \pi_0 (1 + 1.5 + 0.5) = 1$

$\Rightarrow \pi_0 (3) = 1$.

$\Rightarrow \boxed{\pi_0 = 1/3 \Rightarrow \pi_1 = 1/2, \ \pi_2 = \frac{1}{6}.}$

(d) $\pi_i \longrightarrow$ embedded chain (where I jump).
$\tilde{\pi}_i \longrightarrow$ CTMC (how much time spent)

$\therefore$ $\boxed{\text{Time fraction} \propto \text{visit frequency} \times \text{average holding time}}$

In CTMC: holding time in state $i \sim Exp(\mu_i)$
$\mu_i = -q_{ii}$.

so : $E[\text{time in state } i] = 1/\mu_i$

$\therefore \tilde{\pi}_i \propto \pi_i \cdot 1/\mu_i$

$\pi_0 = 1/3$, $\pi_1 = 1/2$, $\pi_2 = 1/6$
$\mu_0 = 4$, $\mu_1 = 6$, $\mu_2 = 4$.

$\therefore \tilde{\pi}_0 \propto \frac{1}{12}$, $\tilde{\pi}_1 \propto \frac{1}{12}$, $\tilde{\pi}_2 \propto \frac{1}{24}$.

Now we normalize.

$$\left(\frac{1}{12}, \frac{1}{12}, \frac{1}{24}\right) \longrightarrow \left(\frac{\frac{1}{12}}{\frac{1}{12}+\frac{1}{12}+\frac{1}{24}}, \frac{\frac{1}{12}}{\frac{1}{12}+\frac{1}{12}+\frac{1}{24}}, \frac{\frac{1}{24}}{\frac{1}{12}+\frac{1}{12}+\frac{1}{24}}\right)$$

$$= \left(\frac{1}{12} \times \frac{24}{5}, \frac{1}{12} \times \frac{24}{5}, \frac{1}{24} \times \frac{24}{5}\right)$$

$$= \left(2/5, 2/5, 1/5\right)$$

$\tilde{\pi}_0 = 2/5$, $\tilde{\pi}_1 = 2/5$, $\tilde{\pi}_2 = 1/5$.

Verified $\rightarrow$ $\tilde{\pi}_0, \tilde{\pi}_1, \tilde{\pi}_2$ values satisfy the condition.

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2) (a) let n be the number of customers at a time.

This includes the one being served, plus everyone waiting.

So states → 0, 1, 2, 3, 4 . . . . .

Arrival (birth): "increase in n
Departure (death): decrease in n.
→ Service completion (1)
→ Abandonment (2)

Departure = (1) + (2)
= 12 + 10(n-1).
Arrival = 10.

Now we have a birth - death process:
$n \to n+1$ [rate = 10],
$n \to n-1$ [rate = $12 + 10(n-1)$].

For ANY CTMC:
(i) off diagonal entries = transition rate
(ii) diagonal = -ve of total outgoing rate.

For state 0: $q_{01} = 10$, $q_{00} = -10$.
" " 1 : $q_{12} = 10$, $q_{10} = 12$, $q_{11} = -22$.
" " 2 : $q_{23} = 10$, $q_{21} = 22$, $q_{22} = -32$.
" " 3 : $q_{34} = 10$, $q_{32} = 32$, $q_{33} = -42$.

$$Q = \begin{bmatrix} -10 & 10 & 0 & 0 & \cdots \\ 12 & -22 & 10 & 0 & \cdots \\ 0 & 22 & -32 & 10 & \cdots \\ 0 & 0 & 32 & -42 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \\ \vdots & \vdots & \vdots & \vdots & \end{bmatrix}$$

(b) using local balance:

$\pi_{n-1} \cdot \lambda_{n-1} = \pi_n \cdot \mu_n$.

$\pi_n = \pi_{n-1} \cdot \frac{\lambda_{n-1}}{\mu_n}$.

$\pi_n = \pi_{n-1} \cdot \frac{10}{12+10(n-1)}$

Also $\sum_i \pi_i = 1$.

$\therefore \pi_1 = \pi_0 \cdot \frac{10}{12}$

$\pi_2 = \pi_1 \cdot \frac{10}{22} = \pi_0 \cdot \frac{10}{12} \cdot \frac{10}{22}$.

$\pi_3 = \pi_2 \cdot \frac{10}{32} = \pi_0 \cdot \frac{10}{32} \cdot \frac{10}{22} \cdot \frac{10}{12}$.

$\pi_4 = \pi_3 \cdot \frac{10}{42} = \pi_0 \cdot \frac{10}{42} \cdot \frac{10}{32} \cdot \frac{10}{22} \cdot \frac{10}{12}$.

$\pi_0 + \pi_1 + \pi_2 \ldots \ldots = 1 \Rightarrow \pi_0 \left(1 + \frac{10^2}{12 \cdot 22} + \frac{10^3}{32 \times 22 \cdot 12} \ldots\right) = 1$.

$\therefore \pi_0 = 1 + \sum_{n=1}^{\infty} \prod_{k=1}^{n} \frac{10}{12+10(k-1)} \rightarrow S$.

$S = 1 + \sum_{n=1}^{\infty} \prod_{k=1}^{n} \frac{10}{12+10(k-1)} = 1 + \sum_{n=1}^{\infty} \prod_{k=1}^{n} \frac{1}{k + 1/5}$

~~$= 1 + \sum_{n=1}^{\infty} \frac{10^n}{\prod 12+10(k-1)}$~~

$\therefore S = 1 + \sum_{n=1}^{\infty} \prod_{k=1}^{n} \frac{1}{k + 1/5} \approx 2.365$.

$\therefore \boxed{\pi_0} = \frac{1}{2.365} = \boxed{0.4228}$ All other π_i can be found from π_0.

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(c) Arrival (New) sees system in state n with probability (stationary), π_n.

If $n = 0$, NO ABANDONMENT possible.

If $n = m$, there are m people ahead of you. If you wait, $(m-1)$ event is given by rate $= 12 + 10(m-1) = 10m + 2$. If you abandon then the rate is 10, we have already done this before.

$$P(\text{survive}) = \frac{10n+2}{10n+2+10} = \frac{10m-2}{10m+12}.$$

$$\therefore P(\text{served} \mid n) = \prod_{m=1}^{n} \frac{10m-2}{10m+12} \Rightarrow \prod_{m=1}^{n} \frac{10(m-1)+12}{10m+12}.$$

$$\therefore P(\text{served} \mid n) = \frac{12}{\cancel{22}} \times \frac{\cancel{22}}{\cancel{32}} \times \frac{\cancel{32}}{\cancel{42}} \times \dots \frac{10\cancel{n-2}}{10n+12}$$

$$= \frac{12}{10n+12}.$$

$$\therefore P(\text{abandon} \mid n) = 1 - \frac{12}{10n+12}$$

$$= \frac{10n + \cancel{12} - \cancel{12}}{10n+12} = \frac{n}{n+1.2}$$

Removing conditioning.

$$\therefore P(\text{abandon}) = \sum_{n=0}^{\infty} \pi_n \left(\frac{10n}{10n+12}\right).$$

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(d) $E[N] = \sum_{n=0}^{\infty} n \cdot \pi_n$.

$= \sum_{n=1}^{\infty} n \cdot \pi_0 \cdot \prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k} = 1 \cdot \pi_1 + 2 \cdot \pi_2 + 3\pi_3 + 4 \cdot \pi_4 + 5\pi_5 \cdots$

$\pi_1 \approx 0.3523$,
$\pi_2 \approx 0.1602$
$\pi_3 \approx 0.0501$,
$\pi_4 \approx 0.0119$,
$\pi_5 \approx 0.0023$

$E[N] = 0.3523 + 2 \times 0.1602 + 3 \times 0.0501 + 4 \times 0.0119 + 5 \times 0.0023 + \cdots$

$E[N] \approx 0.8846$.

Expected number of customers < 1 !

(e) $\mu = 12$.

throughput = $12 \times P$(server is busy)

$P(\text{busy}) = 1 - \pi_0$.

$\therefore$ throughput $= 12 \times (1 - \pi_0)$.

$\pi_0 \approx 0.4228$.
$1 - \pi_0 = 0.5772$

$\therefore$ Throughput $= 12 \times 0.5772$
$= \boxed{6.9264}$

(c) We can use the throughput angle from part (e).

Fraction served = $\frac{\text{throughput}}{\text{arrivals}}$

$$P(\text{abandon}) = 1 - \frac{12(1-\pi_0)}{10}$$

$$\approx \boxed{0.3074} \quad (\text{ans}).$$